Functional Standard Work

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Title: RLPM Sustainment

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Authorized By:	Corporate Vice President - Global Engineering and Program Management

Table of Contents

1. Purpose
2. Persons Responsible
3. Responsibilities
4. Process Or Procedure
5. Revision History
6. Related Information
7. Change Management

1 Purpose

1.1 The purpose of this Standard Work is to provide additional Sustainment Phase requirements and details for compliance with GCP-59 RTX Lifecycle Program Management (RLPM).

2 Persons Responsible

2.1 Scope

2.1.1 The "Sustainment" Phase (Figure 1) intent is:

a. To execute the closure of production capability, and end-of-delivery, for top-level systems and/or select sub-systems and parts for a given product line.

b. To establish end-of-sustainment activity transition planning while continuing to provide operational safety initiatives, customer support and aftermarket strategies.

c. To begin planning during the Pursuit-to-Startup Phase. Sustainment activity begins during the Production Phase.

d. To anticipate and respond to changing logistics support, technical requirements and environments to allow continued operation and support within cost, schedule, and risk constraints.

e. A dual nature in that it provides support to the product as designed and fielded as well as adapting or modifying the support system to accommodate anticipated or unanticipated change.

f. A Discontinuation of Support Gate which authorizes closure of the business case and disbanding of Program Management Team (PMT).

g. To document lessons learned and incorporate into the program closeout registry for review by future projects.

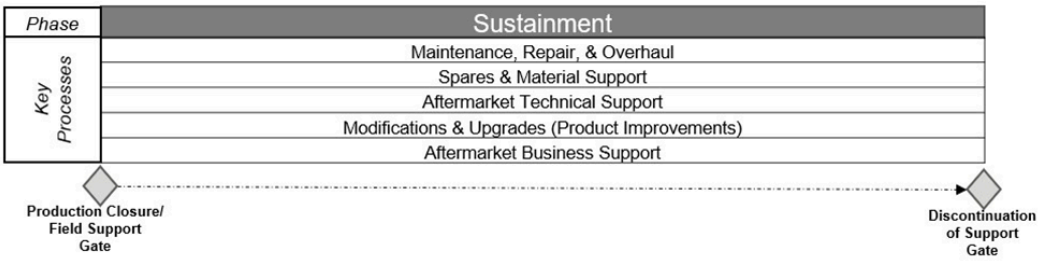

Figure 1: Sustainment Phase Framework

2.1.2 Gates and Milestone Reviews Determination

a. No RTX-required Gates within the Sustainment Phase

i. Production Closure/Field Support gate precedes Sustainment Phase

ii. Discontinuation of Support Gate ceases all program efforts and ends Sustainment Phase

b. Milestone Review prior to support the Sustainment Phase

i. Sustainment Readiness Review

c. Regular Cadence Reviews occur during the Production and Sustainment Phases

i. Applicable broadly to support/sustainment related efforts.

ii. Initiated during Production Phase as aligned with initial product/service deliveries.

iii. Occur at a regular interval until Discontinuation of Support Gate is complete.

2.2 Applicability

2.2.1 This standard work applies to all:

a. Organizations within RTX, including its major Business segments (each, a "**Business**"), divisions, and direct and indirect subsidiaries;

b. Majority-owned or controlled joint ventures and other majority-owned or controlled entities and operations; and

c. Directors, officers, employees, contractors, and any other persons who work for these organizations.

2.2.2 Each Business must implement the provision of this standard work within 6 months following the initial effective date. "Implement" in this context means the Business specific program lifecycle management policies are updated and released in compliance to RLPM.

2.2.3 This standard work applies throughout the RLPM Sustainment Phase. See Exhibit A of GCP-59.

2.3 Persons Affected

2.3.1 This standard work applies to all stakeholders listed in the Responsible, Accountable, Support, and Inform (RASI) Matrix. See Exhibit A.

2.4 Governance

2.4.1 See FSW-TGE-0, Governance of Technology & Global Engineering Documents for governance of this FSW.

3 Responsibilities

3.1 The responsibility assignment matrix inclusive of Responsible, Accountable, Support, and Inform represents responsibilities for activities. For a successful Lifecycle Program Management, each RTX function plays a role in meeting a requirement. See Exhibit A for the RASI that represents functional assignment recommendations.

3.2 In some cases, individuals within certain functions may perform multiple roles.

4 Process Or Procedure

4.1 Requirements

4.1.1 Key Processes Associated with Sustainment. The Aftermarket/Customer Support Lead must include the appropriate processes below as part of the Sustainment Plan. See Figure 2 for the connections associated.

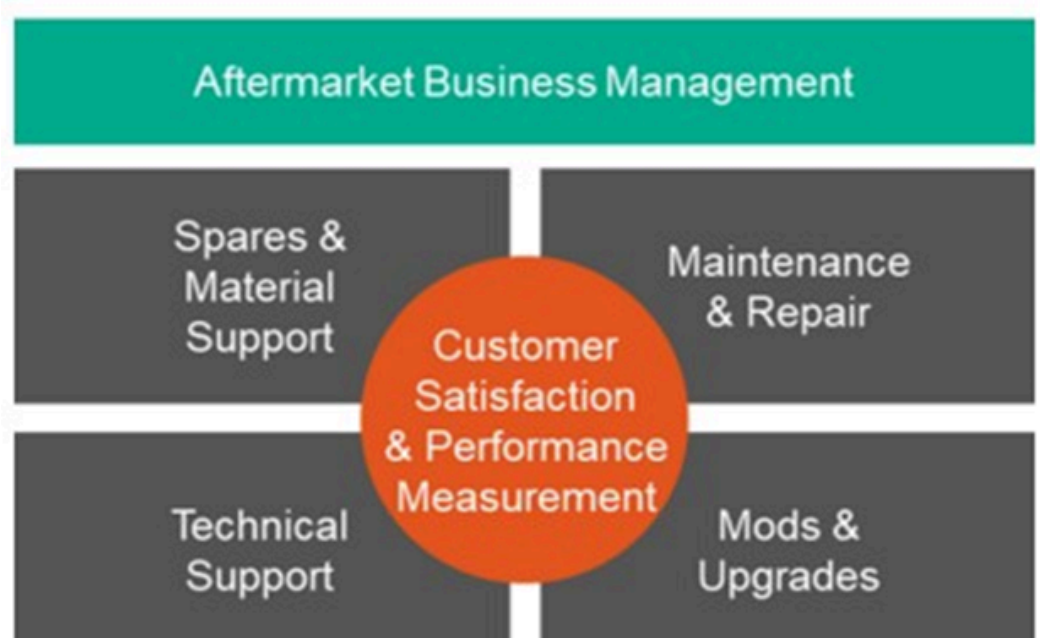

Figure 2: Key Processes View

a. Maintenance, Repair, & Overhaul
    i. Establish Repair and Overhaul Capabilities *
    ii. Maintenance Strategy Planning
    iii. Maintaining Repair and Overhaul Capabilities
    iv. Repair Management
b. Spares and Materials Support
    i. Entry into Service (EIS) *
    ii. Sales Inventory and Operations Planning (SIOP)
    iii. Provisioning
    iv. Inventory Management
    v. Rotable Pools
    vi. Spares Order Administration and Management
c. Aftermarket Technical Support
    i. Product Reliability Management
    ii. Technical Publications
    iii. Operator Field Service Support
    iv. Customer Technical Case Management
    v. Field Technical Resolution Process
    vi. Training
    vii. Ground Support/Test Equipment
d. Modifications and Upgrades
    i. Improvement Identification *
    ii. Improvement Business Case Study *

iii. Modifications and Upgrades Strategy Planning

e. Aftermarket Business Support

i. Sales & Business Development *

ii. Warranty Administration

iii. Market Intelligence Gathering *

iv. Strategy Planning

v. Program Sustainment Plan Development

vi. Contract Management

vii. Performance Monitoring/Contract Compliance

viii. Customer Management

ix. Ao/Flight Hour/Availability Agreement Support

f. Customer Satisfaction/Performance Measurement

i. Quality

ii. Cost

iii. Schedule/Delivery

iv. Safety

v. Customer specific measures (for example, Scorecards)
NOTE: * represents "*Sustainment activity performed during the Production Phase driven by the Sustainment Plan".*
All other text represents "Used during Production and Sustainment Phases"

NOTE: For sites that do not have service and support facilities, that also engage in activities requiring oversight by NASA – National Aeronautics and Space Administration (NASA), Federal Aviation Administration (FAA), European Aviation Safety Agency (EASA), or another governing body, this site may use the same standard processes, procedures, and/or metrics defined in the "Production" Phase during the "Sustainment" Phase.

4.1.2 Required Phase Entrance Criteria (Production Closure/Field Support Gate)

a. Performance of a contractual obligation assessment to determine customer impact.

b. A plan for repair inventory, as required/predicted.

c. An estimate to the end of Sustainment Phase.

d. The cost/benefit analysis for ending production.

e. A communication plan including discontinuation notice for internal and external impacted parties.

f. Updates to the Business, Program, Development, Production, and Sustainment Plans.

g. Updates to the Global Trade export / import plan (if required)

h. Defined Cadence Reviews and key performance indicators

i. Customer/Market schedule and sustainment expectations

j. End-of-Production plan, including resource plan and asset/property transition

k. Product/Services safety requirements plan

l. Program risk management plan (documented, communicated and mitigation plans in place)

m. Post-production aftermarket supportability plan

n. Inventory strategy (for example, spare or lease pool engines/product, parts)

o. Post-certification engineering plan and targeted initiatives, as applicable

p. Improvement plan based on customer/airframer marketing feedback analysis, as required

4.1.3 Required Phase Exit Criteria (Discontinuation of Support Gate)

a. Completion of contractual obligations and customer impacts review

b. A communication plan including discontinuation notice for internal and external impacted parties

c. The cost/benefit analysis for ending sustainment

d. Market analysis/Financial analysis that supports Program exit

e. Approved risk management plans, if necessary

f. Consensus on program exit strategy

g. Consensus on fleet retirement plan with level one schedule

h. Intellectual property provisions with approval by Legal

i. Consensus on trigger condition points through PMT disbanding

j. Gate Approval Process/Authority for Discontinuation of Support Gate

i. Representation from Program Manager, Aftermarket/Customer Support Lead, and Finance Lead must attend and provide approval.

ii. The Business may add other functional leads

iii. Authorization entity as determined by Business DOA.

k. A Global Trade review conducted and closure strategy for the export / import plan, if required

4.1.4 Milestone Review

a. Review Attendance. The Program and Sustainment Plans define who needs to attend reviews.

b. Sustainment Readiness Review

i. Description: Performed in conjunction with Pre-Production Readiness Review, or prior to delivery of first product or service to customer, to ensure released production orders (internal or external) have an updated and actionable Sustainment Plan and adequate resource availability to execute the Sustainment Plan.

ii. Mandatory Criteria

1. Conduct of Pre-Production Readiness Review

2. Conduct prior to first product or service delivered to customer

3. An update and/or review of the Sustainment Plan and readiness for program team execution prior to delivery of first product or service

a. Verification of the program overview, including planned customers and "where-used"

b. Maintenance, Repair, & Overhaul capabilities are in place or have a funded plan to be in place

c. Schedule of Spares & Material Support exists and are in place or have a funded plan to be in place

d. Aftermarket Technical Support artifacts, training and equipment are in place or have a funded plan to be in place

e. Aftermarket Business Support including warranty administration and customer management

4. Facility resources, including Supply Chain resources if applicable, are in place (for example, locations, equipment, manuals, repair, or training)

5. Staffing resources in place

6. Customer Satisfaction/Performance Measurement metrics established including Cadence Review timing and audience

4.1.5 Cadence Reviews within the Sustainment Phase

NOTE: Modifications and Upgrades, Aftermarket Business Support, and Customer Satisfaction/Performance Measurement processes may have Cadence Reviews per Business requirements.

a. Maintenance Repair & Overhaul (MRO)

i. Content per the approved Sustainment Plan and cost of MRO support

ii. Frequency as defined by the Sustainment Plan

iii. Review Representation

1. Representation from Program Manager, Aftermarket/Customer Support Lead, Engineering Lead, Operations Lead, Supply Chain Lead, Legal/Contracts/Compliance Lead, and Finance Lead must attend.

2. The Business may add other functional leads.

3. Authorization entity is not applicable – complete actions as required.

b. Spares & Material Support

i. Content per the approved Sustainment Plan and cost of Spares and Materials support.

ii. Frequency as defined by the Sustainment Plan.

iii. Review Representation

1. Representation from Program Manager, Aftermarket/Customer Support Lead, Engineering Lead, Operations Lead, Supply Chain Lead, Quality Assurance Lead, Legal/Contracts/Compliance Lead, and Finance Lead must attend.

2. The Business may add other functional leads.

3. Authorization Entity is not applicable – complete actions as required.

c. Aftermarket Technical Support

i. Content per the approved Sustainment Plan and cost of aftermarket technical support.

ii. Frequency as defined by the Sustainment Plan.

iii. Review Representation

1. Representation from Program Manager, Aftermarket/Customer Support Lead, Engineering Lead, Supply Chain Lead, and Finance Lead must attend.
2. The Business may add other functional leads.
3. Authorization Entity is not applicable – complete actions as required.

5 Revision History

Rev #	Date	Content Owner	Author	Details
1.0	10.1.2024	M. Delcheccolo	C. Oban	Initial Release

6 Related Information

6.1 Definitions

6.1.1 See GCP-59 for definitions of additional terms used within this FSW.

6.1.2 **Rotable Pools** - The Business holds inventory for rotation to a customer so a failed product can continue operations with minimal interruption. Upon return of the customer product, the company repairs, re-qualifies, and returns the inventory to the pool. Use may include an exchange for a longlead time repair to expedite the build process. Maintaining pools allows the company to increase throughput and avoid late delivery penalties.

6.2 Acronyms

6.2.1 **DOA** - Delegation of Authority

6.2.2 **EASA** – European Aviation Safety Agency

6.2.3 **EIS** - Entry into Service

6.2.4 **FAA** – Federal Aviation Administration

6.2.5 **MRO** - Maintenance Repair & Overhaul

6.2.6 **NASA** – National Aeronautics and Space Administration

6.2.7 **PMT** - Program Management Team

6.2.8 **RASI** - Responsible, Accountable, Support, and Inform

6.2.9 **RLPM** - RTX Lifecycle Program Management

6.2.10 **SIOP** - Sales Inventory and Operations Planning

6.3 Exhibits

6.3.1 Exhibit A - Responsibility Assignment Matrix (RASI)

Based on roles and responsibilities assigned within the Business, the responsible stakeholder may vary from the RASI matrix.

Stakeholders (Role) / Phase or Key Deliverables	Program Manager	Strategy	Business Development/Capture	Engineering	Operations	Supply Chain	Aftermarket/Customer Support	Quality Assurance	Legal/Contracts/Compliance	Finance	Other functional representatives, as required
Sustainment Plan	A	I	S	S	S	S	**R**	S	S	S	I
Key Processes											
Maintenance, Repair, & Overhaul	A	I	I	S	S	S	**R**	S	I	I	I
Spares and Material Support	A	I	I	S	S	S	**R**	S	S	I	I
Aftermarket Technical Support	A	I	I	S	I	S	**R**	I	I	I	I
Modifications and Upgrades	A	I	I	S	S	S	**R**	I	S	I	I
Aftermarket Business Support	A	I	S	S	S	S	**R**	S	S	S	I
Customer Satisfaction and Performance Measurement	A	I	I	I	I	I	**R**	I	I	I	I
Risks and Opportunities Assessment	A	I	S	S	S	S	**R**	S	S	S	I

Legend

a. Responsible (R): Obligation to complete the deliverable or task

b. Accountable (A): Has final decision-making authority and accountability for completion of the deliverable or task

c. Support (S): Provides the support, assistance, and consultation for those responsible for the deliverable or task

d. Informed (I): Informed after a decision or action

6.4 Links to reference material

6.4.1 GCP-59 RTX Lifecycle Program Management

6.4.2 FSW-TGE-0 Governance of Technology & Global Engineering Documents

6.4.3 FSW-TGE-RLPM-01 RLPM Pursuit-to-Startup

6.4.4 FSW-TGE-RLPM-02 RLPM Development

6.4.5 FSW-OPSC-RLPM-03 RLPM Production

7 Change Management

7.1 Communication, awareness and access to the FSW are key to success

7.1.1 Communicate and train employees to perform tasks as defined in the FSW

7.1.2 Post the FSW in a location accessible by all impacted persons (examples include SharePoint Online, Teams, and ePolicy)

7.1.3 Periodically review and update the document to incorporate improvements.